

RESEARCH ARTICLE Hot Paper

Crown Ether Rotaxane-Induced Construction of COF Membranes With Recognition Channels for High-Efficiency Ion Sieving

 Liang Ren¹ | Jinrui Song¹ | Nan Zhang¹ | Yafei Guo¹ | Nelson Belzile² | Tianlong Deng¹ 
¹State Key Laboratory of Bio-based Fiber Materials, Key Laboratory of Marine Resource Chemistry and Food Technology in Ministry of Education, Tianjin University of Science and Technology, Tianjin, P. R. China | ²School of Natural Sciences, Laurentian University, Ontario, Canada

 Correspondence: Tianlong Deng (tldeng@tust.edu.cn)

Received: 17 November 2025 | Revised: 21 January 2026 | Accepted: 22 January 2026

Keywords: biomimetic membrane | COF nanosheet | ion recognition | ion sieving | rotaxane chemistry

ABSTRACT

Biological ion channels can achieve high ion discrimination through the synergy between pore structure and microenvironment. However, constructing biomimetic ion-sieving membranes with precise recognition capabilities for target ions remains challenging. Herein, we propose a rotaxane-induced stacking method for constructing COF membranes integrated with ion-recognition sub-nanoscale channels. The dibenzo-crown ether rotaxane COF (CRCOF) nanosheets are fabricated and subsequently stacked into membranes. Driven by π - π interactions between rotaxane moieties and CRCOF nanosheets, as well as the specific ion-recognition ability of rotaxanes, the nanosheets undergo oriented stacking, yielding well-defined sub-nanoscale channels equipped with recognition sites. The angstrom-scale pore size and specific binding channels synergistically enhance selectivity and minimize transport energy penalties of target ions. By modulating the ion recognition capability of channels, the obtained CRCOF membrane demonstrates an exceptional Li^+ permeation rate of $0.04 \text{ mol m}^{-2} \text{ h}^{-1}$ (approximately five times higher than reported polymer membranes) and high Li^+ selectivity ($\text{Li}^+/\text{Mg}^{2+}$ selectivity of 315 and Li^+/Na^+ selectivity of 12) in a mixture solution. This work provides a new avenue for the accurate construction of biomimetic ion-sieving membranes and offers new insights into the mechanisms of high-efficiency ion separation in sub-nanoscale confined recognition channels.

1 | Introduction

In nature, biological ion channels can achieve efficient recognition and selective transport of specific ions through the fine-tuned matching of pore sizes and the multifunctional synergy of microenvironments (e.g., hydrogen bonding, coordination, and electrostatic interactions) [1, 2]. For instance, the KcsA channel can precisely distinguish between K^+ and Na^+ , achieving a selectivity ratio exceeding 10^4 even when the difference in their ionic radii is less than $0.3 \text{ \AA}$ [3, 4]. Its synergistic mechanism based on the regulation of dehydration energy barriers

and pore size sieving has become a classic paradigm in ion selectivity research. Such high selectivity derived from structure-microenvironment synergy provides an ideal biomimetic prototype for addressing ion separation challenges in the fields of resources, energy, and environment [5–8]. As a critical resource in the new energy industry, Li^+ requires efficient separation from coexisting ions (e.g., Na^+ , Mg^{2+}), which is particularly imperative [9, 10]. The existing lithium extraction technologies often require coupling with a multi-step, time-consuming, and energy-intensive pretreatment process, severely undermining the efficiency of membrane-based lithium extraction [11–13]. The

development of artificial counterparts with recognition capabilities similar to those of biological channels has become a decisive factor in promoting the efficient extraction of lithium resources.

In recent years, biomimetic ion separation membranes have been developed using materials such as porous organic polymers (POPs), metal-organic frameworks (MOFs), and covalent organic frameworks (COFs) [14–16]. It has been found that when the membrane pore size is smaller than the hydrated ion diameter, ions undergo complete or partial dehydration upon entering the pores, and ion separation can be achieved based on differences in dehydration difficulty [17, 18]. In light of this mechanism, the currently fabricated membranes can achieve high-efficiency separation of Li^+ and multivalent cations (e.g., Ca^{2+} , Mg^{2+}). Although Li^+ has a smaller bare ion diameter (1.2 Å), Na^+ and K^+ possess lower dehydration energies ($< 400 \text{ kJ mol}^{-1}$) and are more prone to partial dehydration, accordingly enabling transmembrane transport [19, 20]. Thus, it is challenging to achieve efficient Li^+ separation from other alkali metal ions relying solely on the size sieving effect. Recent studies have demonstrated that introducing target ion recognition sites (e.g., charged groups, chelating groups, specific recognition groups, etc.) into membrane pores can compensate for the dehydration energy of target ions when they enter the pores [21–23]. The stronger the binding ability of the recognition sites, the easier it is for target ions to enter the pores; however, excessively strong binding ability is detrimental to ion transport within the pores [24, 25]. Therefore, the precise introduction of recognition sites with appropriate binding strength into sub-nanometer channels constitutes an effective strategy for constructing high-performance biomimetic ion separation membrane materials.

Two Dimensional COFs, as a class of porous crystalline materials formed by covalently linking planar building monomers, feature both designable pore structures and channel environments [26–28]. Using COF nanosheets with high crystallinity and large aspect ratios (lateral size $> 10 \mu\text{m}$, thickness $< 10 \text{ nm}$) as building blocks, continuous and defect-free stacked membranes can be rapidly formed via vacuum-assisted self-assembly, spin-coating, or drop-casting, exhibiting unique prospects in the construction of biomimetic ion separation membranes [29–31]. However, limited by the fact that the intrinsic pore size of COF materials is generally larger than 1 nm, the difficulty in constructing functionalized COF nanosheets, and the poor controllability of nanosheet stacking processes, it is challenging to precisely construct biomimetic ion separation membranes with uniform sub-nanometer channels, tunable recognition sites, and 1D vertical orientation. Recent research has shown that functional side groups on building monomers can enable the synergistic regulation of pore structure and properties of COFs. For instance, chain-like side groups with different steric hindrances can affect the crystal stacking energy of COFs, thereby provoking their nanosheets to stack in different modes; additionally, they can change the hydrophobic properties of the material [32]. Likewise, non-planar side groups can trigger interlayer distortion of COFs, facilitating angstrom-scale regulation of membrane pore size and tuning their mechanical flexibility [33, 34]. Given this, fabricating COF nanosheets decorated with functional side groups exhibits great potential in the construction of 1D sub-nanoscale recognition channel membranes.

Herein, we propose a dibenzo-crown ether rotaxane COF (CRCOF) membrane bearing angstrom-scale ion-recognition channels via a rotaxane-induced building method (Figure 1). Crown ethers, as a kind of flexible ring molecule, are endowed with guest molecule binding capability, ion recognition ability, and tunable pore sizes. They form rotaxane COF nanosheets through host-guest interactions with building monomers, thus inducing the nanosheets to stack and assemble into ordered 1D sub-nanometer channel membranes. Moreover, the oxygen atoms in crown ethers constitute ion recognition sites with a guanidyl nitrogen atom in the COF backbone. By adjusting the dibenzo-crown ether (DCE) content and type, as well as membrane thickness, the recognition selectivity for different ions can be precisely modulated. The synergistic optimization of sub-nanometer channels and recognition sites in the membrane enables efficient dehydration and rapid transmembrane transport of target ions, laying the foundation for direct lithium extraction from complex salt solutions.

2 | Results and Discussion

The CRCOF membranes were formed through assembly of nanosheets induced by rotaxanes (Figure 2a). Due to TbTG_{Cl} COF having a sub-1 nm pore size, a positively charged pore wall, and the ease of forming layered materials, we selected it as a model case. For CRCOF nanosheet synthesis, DCE macrocycles were first threaded onto TG_{Cl} monomers via host-guest interactions, generating pseudorotaxane structures. The CRCOF nanosheets were then obtained via a single-phase method (Figures S1 and S2) [35], followed by membrane fabrication through pressure-assisted assembly. The benzene rings on the rotaxane would exert π - π induced effects during membrane forming. The obtained membranes not only had vertically aligned 1D nanochannels, but also developed reduced apertures because of the shielding of rotaxanes. Graded rotaxane functionalization was achieved through stoichiometric control of crown ether addition (DCE/ TG_{Cl} molar ratios are 0:3, 1:3, 2:3, and 3:3). The membranes were designated as CRCOF-X, where CRCOF-0 represents the pristine membrane without rotaxane incorporation.

The morphology of the obtained CRCOF nanosheets was characterized by scanning electron microscopy (SEM) and transmission electron microscopy (TEM). The SEM results demonstrated that nanosheet morphology remained consistent after the introduction of crown ether rotaxanes (Figures 2b and S3). TEM analysis revealed that the nanosheets exhibit an ultrathin thickness and large lateral dimensions. The HRTEM images showed that the nanosheets had high crystallinity (Figures 2c, S4, and S5). In the selected area electron diffraction (SAED) pattern, the CRCOF-2 nanosheets had clear diffraction spots. The mapping diagram showed that the proportion of oxygen atoms increased, which also corroborated the successful introduction of crown ether rotaxane into the nanosheets (Figures S6 and S7). Finally, atomic force microscopy (AFM) was employed to characterize the thickness of nanosheets (Figure 2d). The measurement results revealed that the thickness of a single layer was approximately 5 nm, with an aspect ratio exceeding 2000.

CRCOF nanosheets can form composite membranes with adjustable separation layer thickness on porous substrates

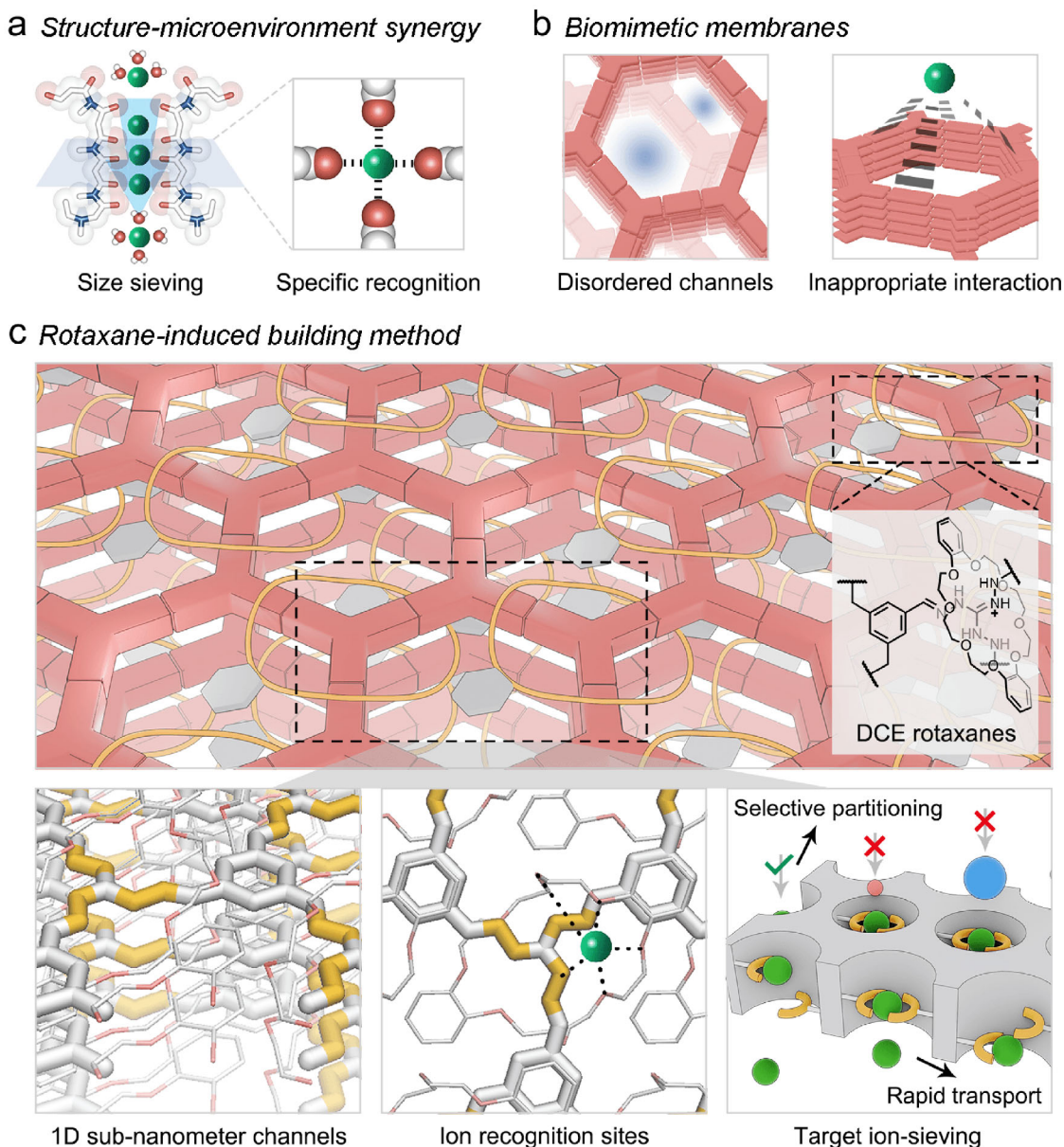


FIGURE 1 | Schematic illustration of biological ion channels and CRCOF membrane construction. (a) A biological ion channel with a subtle pore structure and recognizable channel environment. (b) Existing issues of lamellar stacking biomimetic membranes. (c) Construction of an efficient ion-sieving CRCOF membrane with ordered ion recognition channels via a rotaxane-induced method.

through vacuum filtration, and there was a linear relationship between their thickness and the volume of nanosheet dispersion (Figure S8). After stripping, the CRCOF membranes can be transferred to any substrates, and the membranes with a thickness of more than a micrometer can exist in a self-standing form, which demonstrates the robustness of COF membranes (Figure S9). The morphology of CRCOF membranes on the AAO substrate was characterized by SEM (Figures 2e,f and S10). The images showed that the membranes had a smooth surface, and the cross-sectional view showed that the separation layer on the porous substrate had a layered stacking morphology. The AFM further characterized the surface roughness and thickness of the membrane (Figures 2g and S11), with its R_a value less than 30 nm, and the thickness measurement was consistent with that of the SEM measurement.

The chemical structure of the self-standing membrane was characterized by Fourier transform infrared spectroscopy (FTIR). The characteristic peak at 1625 cm^{-1} was C=N stretching vibrations [36], which proved the occurrence of Schiff base reaction (Figures 3a and S12). In the CRCOF-2 membrane, the peak of C–O stretching vibration appeared at 1209 cm^{-1} [37], revealing the successful introduction of the crown ether ring. The XPS broad peak also demonstrated that the C/O ratio of CRCOF-2 membrane was lower than that without introducing the crown ether (Figure S13). In ^{13}C magic angle spinning nuclear magnetic resonance (^{13}C MAS NMR) spectra (Figure S14), carbon resonance signal with a chemical shift in 142.3–159.6 ppm was credited to guanidino carbon atoms in COF. And the peak at 40.2 ppm was attributed to the alkyl carbon of the crown ether ring in CRCOF [38]. The above results implied that the crown ether had been

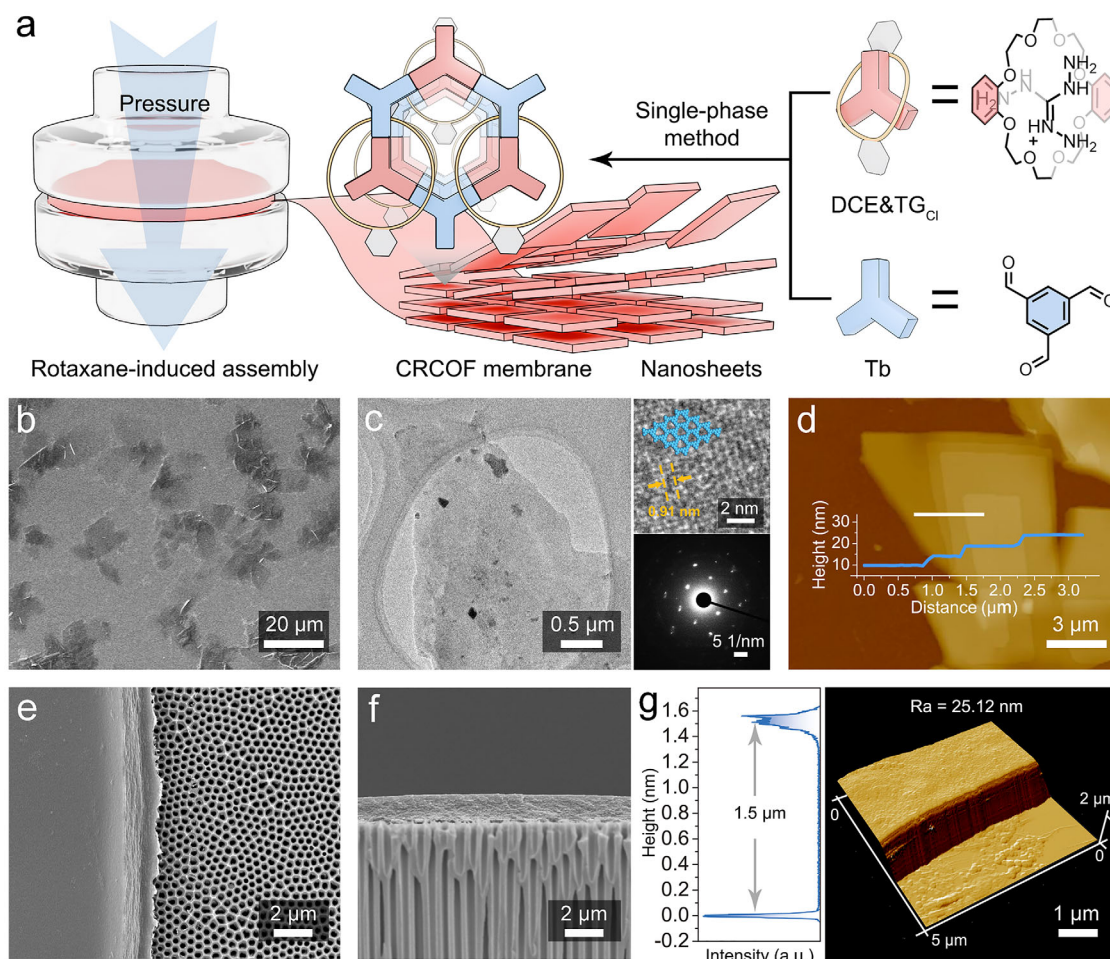


FIGURE 2 | Fabrication of membranes. (a) Schematic diagram of rotaxane-induced CRCOF membrane construction. (b) SEM image of CRCOF-2 nanosheets. (c) TEM image of CRCOF-2 nanosheets. Insert: the HRTEM image along the [001] direction (top) and SAED pattern (bottom) for nanosheets displaying distinct lattice fringes and diffraction spots. (d) AFM topography of CRCOF-2 nanosheets showing a single-layer thickness of ~ 5 nm and an aspect ratio >2000 . (e) Surface and (f) cross-sectional SEM images of CRCOF membranes exhibiting a continuous, defect-free surface and uniform thickness. (g) AFM topography of CRCOF membranes for thickness (~ 1.5 μm) and roughness measurement ($R_a = 25.12$ nm).

successfully combined into the COF. Finally, the thermal stability of the membranes was evaluated by thermogravimetric analysis (TGA), and decomposition occurred at temperatures exceeding 200°C (Figure S15), suggesting the high-temperature application potential of CRCOF membranes.

The lone pairs of electrons on the oxygen atoms in the crown ether can generate an electron-rich cavity [39]. With its content increasing, the zeta potential of CRCOF membranes decreased from 32.2 to 12.3 mV. It was inferred that electrostatic neutralization occurred between the crown ether and TG_{Cl} monomer. The electrostatic potential (ESP) of TG_{Cl} and DCE was simulated to study the interaction (Figure 3b) [40]. The oxygen atoms in the crown ether were negatively charged and guanidine groups in COF showed positive charge, which further proved that electrostatic combination can occur between them. The water contact angle (WCA) measurements were used to characterize the hydrophilicity of membranes. With the increase of DCE content, the WCA increased from 49.1° to 59.7° (Figure S16). This could be assigned to the enhanced hydrophobicity of membranes from the benzene rings introduced by DCE, which facilitated the dehydration of hydrated ions upon entering channels. Due

to the positive charge of undecorated nanosheets, the formation of regular membrane channels will be affected by electrostatic repulsion during the stacking process [41]. However, the addition of crown ether will weaken the charge interaction between the nanosheets. In addition, the π - π interaction between benzene rings in the DCE will induce the AA stacking of the nanosheet to achieve the 1D vertical channels. PXRD showed that the crystallinity was improved after modification and the result matched well with the calculated (Figures 3c, S17–S19, and Table S1).

The orientation of membrane was evaluated by GIWAXS measurements using a synchrotron-based 10 keV x-ray source ($\lambda = 1.2398$ Å). In the GIWAXS pattern of the CRCOF-2 membrane (Figure 3d), the Bragg diffraction peak of the (100) crystal plane (corresponding to the pore structure) at $q_{xy} = 8.56$ nm $^{-1}$ exhibited a significantly stronger in-plane reflection signal than the out-of-plane reflection. Meanwhile, the diffraction peak of the (001) crystal plane (corresponding to π - π stacking) at $q_z = 23.2$ nm $^{-1}$ showed a notably stronger out-of-plane reflection signal compared to the in-plane reflection. These features indicate that CRCOF nanosheets are arranged parallel to each other, with the pore direction perpendicular to the substrate, demonstrating

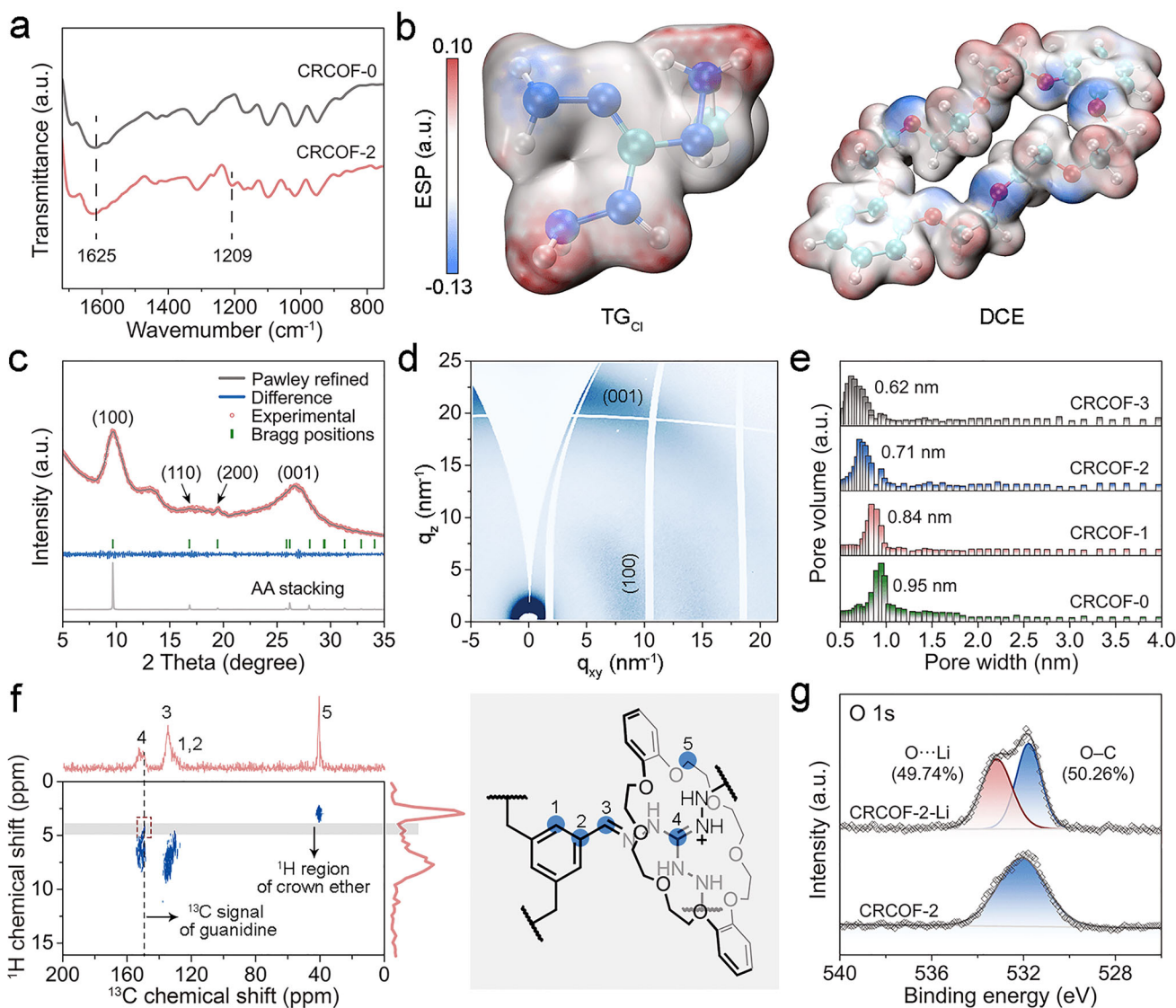


FIGURE 3 | Chemical structure and channel microenvironment of membranes. (a) FTIR spectra of CRCOF-0 and CRCOF-2 membranes. (b) ESP simulation for TG_{Cl} and DCE. (c) PXRD pattern of the experimental (red) and Pawley refined (gray) CRCOF-2 with their refinement difference (blue), and the simulated patterns for eclipsed AA stacking modes. (d) GIWAXS pattern of CRCOF-2 membrane. (e) Pore size distribution curve of CRCOF-X membranes via N_2 adsorption/desorption isotherms. (f) 2D HETCOR spectrum of CRCOF-2 showing a clear correlation spot (red dashed box) between the DCE and the COF. (g) O 1s XPS spectra of CRCOF-2 and CRCOF-2-Li displaying a C-O...Li peak exclusively at 533.2 eV in CRCOF-2-Li, with no such peak detected in CRCOF-2.

a well-defined specific orientation. In contrast, the unmodified COF only exhibited weak orientation (Figure S20), confirming that crown ether modification is the key to inducing the ordered orientation of COF nanosheets. The membranes pore structure was analyzed by N_2 adsorption/desorption isotherms at 77 K and the pore size distribution was calculated by the non-local density functional theory (NLDFT) model (Figures 3e and S21). The results revealed that with the increase of rotaxane content, the pore size decreased from 0.95 to 0.62 nm, which indicated that the shielding effect of rotaxanes reduced the pore size of membranes.

To confirm the periodic distribution of crown ether recognition sites within 1D channels, the solid-state 2D 1H - ^{13}C heteronuclear correlation NMR (HETCOR NMR) measurement was

carried out (Figure 3f). The 2D HETCOR spectrum demonstrated that the methylene hydrogens of the crown ether ring ($-O-CH_2-CH_2-O-$) exhibited heteronuclear correlation peaks with the imino carbons of the COF nanosheets (carbons on the guanidine building monomers). This indicates an extremely short spatial distance (≤ 0.5 nm) between the crown ether rings and the COF matrix [42, 43], implying the occurrence of crown ether ring intercalation behavior rather than simple surface adsorption. These findings indirectly corroborate the existence of periodic recognition sites on the COF. The recognition of Li^+ in CRCOF-2 membrane channels was investigated using XPS. The O 1s spectrum displayed a C-O peak at 531.8 eV for CRCOF-2-Li, which was consistent with CRCOF-2 (Figure 3g). Additionally, a characteristic C-O...Li peak was observed at 533.2 eV, in agreement with the characteristic binding energy of C-O groups

in the known Li(12C4)Cl structure [44]. The integral areas ratio of the two peaks was close to 1, and it was inferred that there were about four oxygen atoms coordinating with Li⁺. Through the peak area ratio (1:25) of N...Li (403.2 eV) to C=N (398.8 eV) and N-N (399.5 eV), it can be inferred that about one-third of a nitrogen atom was coordinated with Li⁺ (Figure S22). Finally, the Li 1s spectral analysis confirmed the characteristic peak position at 56.8 eV in CRCOF-2-Li (Figure S23), which aligned with the reported Li 1s binding energy in LiN(SiMe₃)₂·12C4 [45]. The above results demonstrated that the Li⁺ recognition sites in CRCOF-2 channels adopted a five-coordinated structure, consisting of four crown ether oxygen atoms and a partial guanidyl nitrogen.

The selective transport of cations was investigated via a concentration gradient-driven diffusion test. To systematically elucidate the structure-performance relationship, key structural parameters of DCE content, membrane thickness, and DCE type were independently controlled while monitoring performance variations. The corresponding descriptive parameters of recognition sites density, stacking number, and DCE molecular mass were introduced to understand the transport behavior of ions in COF channels with recognition ability at the molecular level.

The effect of recognition sites density on ion transport was studied by tuning the introduced content of DCE. Based on the DCE/TG_{Cl} molar ratio (0:3, 1:3, 2:3, and 3:3), the recognition sites density in CRCOF-1, CRCOF-2, and CRCOF-3 membranes could be represented as 5.5, 11, and 16.5 mmol cm⁻³, respectively (Figure 4a). In the single-salt system, the feed side contained 0.1 M single-salt solution, and the permeated side had an equal volume of DI water as the driving force. As shown in Figures 4b and S24, the CRCOF-0 membrane exhibited a high permeation rate, but the ion selectivity was poor. The permeation rate showed a decrease with increasing DCE content. The selectivity displayed enhancement, and the CRCOF-2 membrane presented optimal Li⁺/Na⁺ (4.23) and Li⁺/Mg²⁺ (46.22) selectivity. This phenomenon can be due to the reduced pore size and increased density of recognition sites. Specifically, as the content of introduced DCE increased, the membrane pore size decreased from 0.98 to 0.62 nm, and the density of recognition sites per unit volume increased by two times. It was noteworthy that the CRCOF-3 membrane with smaller pore sizes and higher recognition site density failed to achieve maximum selectivity. This may be assigned to the overly strong interactions within the channels that hindered the effective transport of Li⁺ through the membrane.

Subsequently, the influence of channel length on ion separation performance was investigated by altering the membrane thickness. By adjusting the number of stacked layers, membranes with varying thicknesses were fabricated and designated as CRCOF-2(a), where a = 500, 1500, 2500, and 5000 nm, corresponding to layer counts of 1500, 4500, 7500, and 15000, respectively (Figure 4c). The cross-sectional SEM images revealed a linear relationship between the stacking number and membrane thickness, and similar smooth morphologies were observed from the surface SEM images (Figure S25). The ion transport energy barriers of CRCOF membranes with different thicknesses were calculated by fitting the temperature-dependent membrane conductivity based on the Arrhenius equation. The results indicated that as the thickness increased, the transport energy barrier for Li⁺ exhibited an increasing trend (Figure S26).

Ion diffusion test revealed that as the membrane thickness increased, the ion permeation rate exhibited attenuation, and the Mg²⁺ permeation rate showed more pronounced decay (Figures 4d and S27). The Li⁺/Na⁺ and Li⁺/Mg²⁺ selectivity increased with membrane thickness until saturation occurred at CRCOF-2(5000). This observation suggested that increasing the channel length enhanced the interaction between sites and ions, hence improving ion selectivity. However, excessively long channels may compromise ion transport efficiency. Considering both the selectivity and permeation rate, the CRCOF-2(1500) membrane was identified as the most promising candidate for effective lithium extraction.

Different types of DCE have diverse cavity sizes and binding affinities for alkali metal ions [46]. By introducing various DCEs (including dibenzo-15-crown-5, dibenzo-18-crown-6, dibenzo-24-crown-8, and dibenzo-30-crown-10), the ion selectivity of membranes could be tuned. The resulting membranes were denoted as CRCOF-2(b), where b represents the total number of atoms in the crown ether skeleton (b = 15, 18, 24, and 30). The corresponding oxygen atom contents at the recognition sites were 2.5, 3, 4, and 5, respectively (Figure 4e). Density functional theory (DFT) calculations were performed to determine the ion binding energies of different DCEs (Figure S28 and Table S2). The results demonstrated that the binding affinity toward Li⁺ improved with the enlargement of the crown ether ring. Moreover, for the same crown ether ring, Li⁺ exhibited a higher binding energy compared with Na⁺ and Mg²⁺. In addition, the increase of atoms in the crown ether would enhance the steric hindrance of the rotaxane. N₂ adsorption/desorption isotherms showed that the pore size of the membrane decreased successively (Figure S29). SEM images displayed that CRCOF membranes had smooth surfaces and similar thicknesses (Figure S30), suggesting that the type of crown ether had little effect on the morphology of membranes. In the permeation tests, the ion permeation rate decreased with the increase of oxygen atoms (Figures 4f and S31). The selectivity of Li⁺/Na⁺ and Li⁺/Mg²⁺ initially increased and then decreased with the rising number of oxygen atoms, and the optimal value appeared at CRCOF-2(24). This manifestation indicated that increasing the crown ether's atom number elevated the transport resistance of channels while simultaneously altering ion selectivity. The dibenzo-24-crown-8 rotaxane exhibited superior Li⁺ recognition capability and appropriate steric hindrance effect, implying the best lithium extraction performance.

In the mixed-salt system, the salt solution consisted of 0.1 M NaCl, 0.1 M LiCl and 0.1 M MgCl₂. The CRCOF membrane exhibited higher Li⁺ selectivity in this system, and the Li⁺/Na⁺ and Li⁺/Mg²⁺ selectivities were enhanced by about 1.8-fold and 5.8-fold compared to the single, respectively. And its performance was superior to that of most reported membranes (Figures S32 and 4g, Table S3). For example, under the same test conditions, polymer membranes with similar Li⁺/Mg²⁺ selectivity exhibited a Li⁺ permeation rate of only 0.01 mol m⁻² h⁻¹, while the membrane prepared in this work achieved a rate as high as 0.04 mol m⁻² h⁻¹. The anomalous phenomenon may be caused by the Li⁺ recognition site. Due to the high recognition ability, Li⁺ with lower hydration energy at the membrane interface was more likely to dehydrate and enter the membrane than Na⁺ and Mg²⁺. The preferential entry of Li⁺ into the subnanochannel strengthened the steric hindrance (channel occupancy effect), which in turn

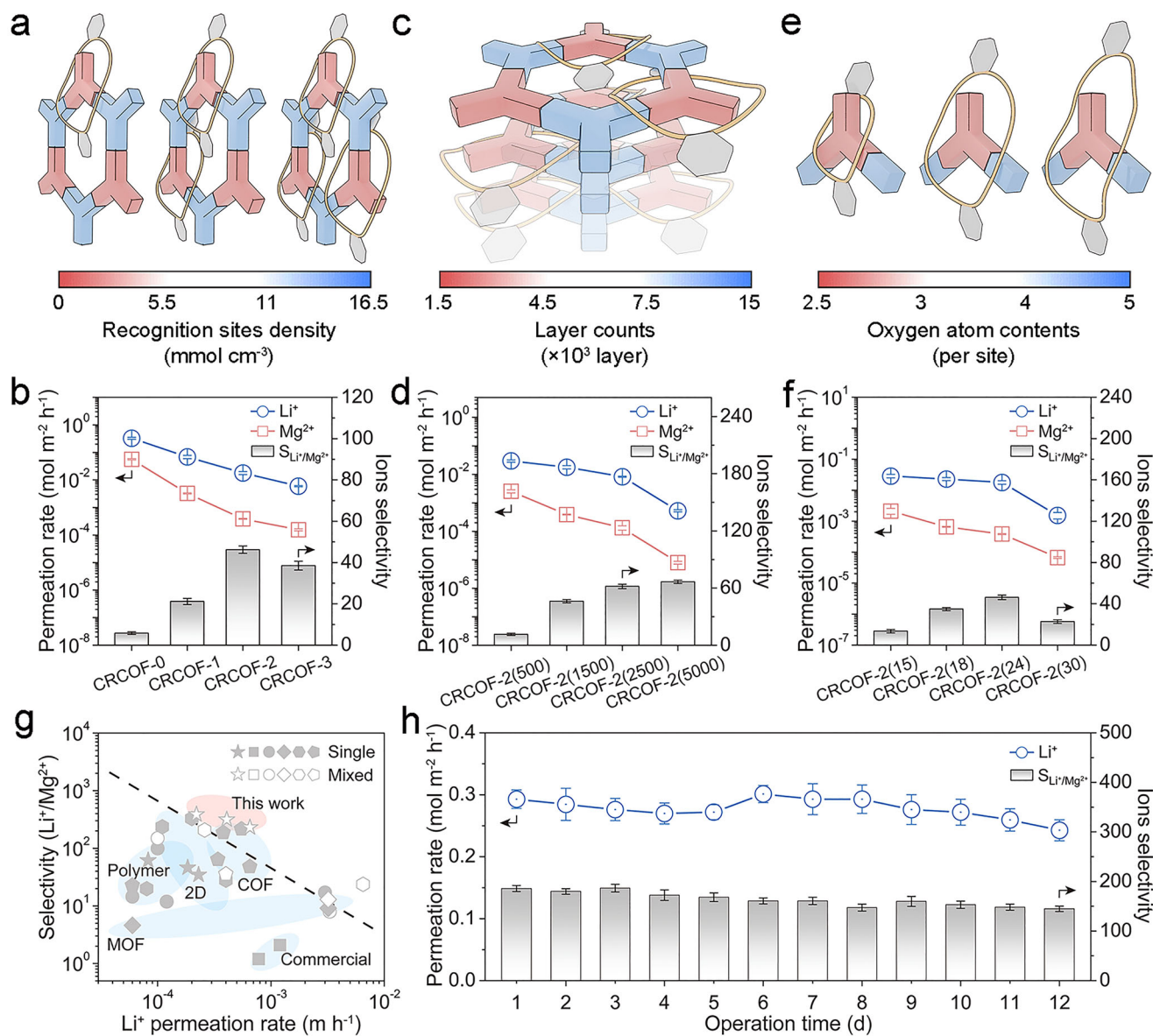


FIGURE 4 | Ion transport properties of CRCOF membranes. Schematic illustration of membrane parameter adjustment: (a) DCE content, (c) membrane thickness, and (e) DCE type. Ion separation performance of (b) CRCOF-X, (d) CRCOF-2(a), and (f) CRCOF-2(b) membranes in a single-salt system. (g) Separation performance comparison with the commercial (Nafion 117 and 212) and reported membranes. (h) The long-term stability of the CRCOF-2 membrane under electrodialysis operation.

significantly impeded Na⁺ and Mg²⁺ transport. In addition, the positively charged sites between adjacent layers can accelerate the Li⁺ migration jump to the next recognition sites, realizing the rapid ion transport in the membranes. However, this effect was not detected without Li⁺, and both Na⁺ and Mg²⁺ can achieve transport through dehydration, resulting in a lower Li⁺ selectivity in the single-salt system. This characteristic holds significant implications for practical lithium extraction from salt lake brines [47]. We simulated the Yiliping brine (Mg²⁺: 0.96 M, Na⁺: 3.48 M, and Li⁺: 0.046 M) and evaluated the Li⁺ extraction performance and long-term operational stability using a home-made electrodialysis device (Figure S33 and Table S4). The Li⁺/Na⁺ molar ratio changed from 0.013 to 22.5, and the Li⁺/Mg²⁺ molar ratio enhanced from 0.048 to 231.2 (Figure S34). Moreover, no significant decline in selectivity was observed after 288 h of operation (Figure 4h), demonstrating the excellent stability of membranes.

To elucidate the ion-sieving mechanism of CRCOF membranes, molecular dynamics (MD) simulations were employed to analyze the behavior of ions during passage across the membrane (Figure S35). The transmembrane process primarily involves ion partitioning and intramembrane diffusion [48]. Ion partitioning refers to the transfer of ions from the bulk solution phase to the membrane surface, whereas intramembrane diffusion denotes the migration of ions within the membranes.

The ion partitioning process primarily involves the dehydration of ions and interactions with membrane channels [49]. To clarify the hydration state of ions when they enter the pores, the transport behavior of water molecules in membrane channels was first investigated. First-principles calculations of water molecule transport rate in the channels revealed that the rate for the CRCOF-0 membrane was 5.89×10^{11} molecules mol⁻¹ s⁻¹, while

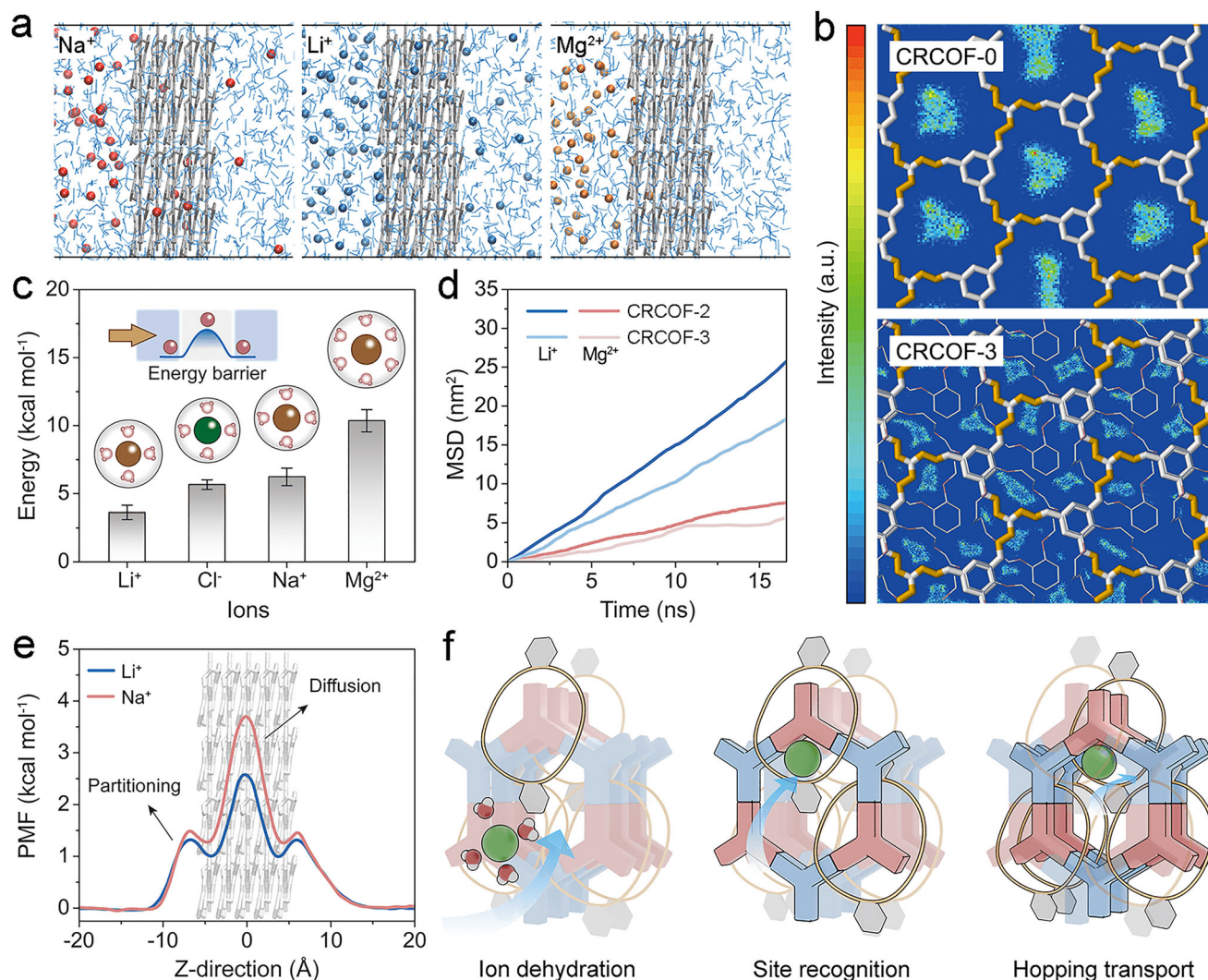


FIGURE 5 | Ion-sieving mechanism in CRCOF membranes. (a) Instantaneous configuration diagrams of ion transport in CRCOF-3 membrane. (b) 2D maps of Li⁺ distribution in different membranes. (c) Interaction energies of ions with CRCOF-2 membrane. (d) MSD of ions in different membranes. (e) PMF of Na⁺ and Li⁺ in CRCOF-2 membrane. (f) Schematic diagram of the target ions transmembrane process in the CRCOF-2 channel.

that for the CRCOF-3 membrane was only 3.65×10^{-15} molecules mol⁻¹ s⁻¹ (Table S5). The latter value was much lower than that of the former, indicating that water molecules can hardly pass through the CRCOF-3 channels. In the water migration experiment, the volume of water reduced in the feed side of the CRCOF-3 membrane was much lower than that of the CRCOF-0 membrane, which was consistent with the above simulation results (Figures S36 and S37). This occurrence was directly related to the introduction of DCE. The formed crown ether rotaxane not only reduced the pore size (from 0.95 to 0.62 nm) and increased the steric hindrance of the channels, but also led to the benzene rings residing on the pore walls, which enhanced the hydrophobicity of the channels. Since water molecules had difficulty passing through the channels of CRCOF-3, and ions usually existed in a hydrated form, ions thus required complete dehydration to enter this channel. In contrast, from the results of the radial distribution functions (RDFs) (Figure S38), it can be seen that ions required only partial dehydration upon entering CRCOF-0 channels.

Dehydrated ions would interact with membrane channels, and when there were binding interactions between ions and the pores, such interactions can compensate for the dehydration energy during ion partitioning; in contrast, when there were repulsive interactions between ions and the pores, the energy barrier in the ion partitioning process would be further increased [50]. Instantaneous configuration diagrams showed that, after a 20 ns simulation, Li⁺ permeated through the pores of CRCOF-3 membrane more easily than Na⁺ and Mg²⁺ (Figure 5a). The simulation results were consistent with the experimental ones, indicating that the bare ion diameter of Li⁺ matched the size of five-coordinate binding sites (approximately 1.7–2.2 Å) [51]. Such specific binding can compensate for the high dehydration energy of Li⁺ relative to that of Na⁺, enabling the selective transport of Li⁺ through the membrane channels (Table S6). Two Dimensional maps showed that Na⁺, Li⁺, and Mg²⁺ can all pass through the CRCOF-0 membrane, with their main distribution in the middle of the channels (Figures 5b, S39, and S40). The order of distribution concentration from high to low was Na⁺ > Li⁺ > Mg²⁺, indi-

cating that the pores were sufficiently large to permit the simultaneous passage of both monovalent and divalent ions, and there was an electrostatic repulsion between the pore walls and the ions. Whereas in the CRCOF-3 membrane, Li^+ exhibited the highest distribution density and tended to be concentrated around the recognition sites. Compared with Li^+ , the distribution densities of Na^+ and Mg^{2+} were significantly reduced in the CRCOF-3 channels. This indicated that the introduction of crown ether sites can enhance the interaction with Li^+ through a specific recognition effect: on one hand, it promoted the enrichment of Li^+ near the recognition sites and reduced the energy barrier for Li^+ to enter the channels; on the other hand, both Na^+ and Mg^{2+} exhibited relatively weak interactions with the crown ether recognition sites, making it difficult for them to bind effectively and enter the channels. This differential interaction with various cations ultimately enabled the membrane to achieve high Li^+ selectivity.

During the intramembrane diffusion of ions, while the introduction of recognition sites can reduce the energy barrier in the ion partitioning stage, an excessively strong binding interaction would restrict the diffusion of ions within the channels, and consequently cause an elevation of the overall transmembrane energy barrier [52]. Calculations of the interaction energies between ions and the membrane revealed that, for ions in CRCOF-2 and CRCOF-3 membranes, the order of interaction energies from highest to lowest was: $\text{Mg}^{2+} > \text{Na}^+ > \text{Cl}^- > \text{Li}^+$ (Figures 5c and S41). This trend was negatively correlated with the order of ion diffusion rates. Although the CRCOF-3 membrane contained more crown ether rotaxane recognition sites, compared with CRCOF-2, the energy barrier that ions needed to overcome when permeating through the CRCOF-3 membrane was larger. This may be attributed to excessively strong binding interactions within the pores. Next, the mean squared displacement (MSD) of ions during transmembrane processes was evaluated, and the ion diffusion coefficients were calculated, as shown in Figures 5d and S42. The results indicated that the ion diffusion in CRCOF-2 and CRCOF-3 membranes had similar behavior, among which the Li^+ in the CRCOF-2 membrane exhibited the largest diffusion coefficient ($D = 2.6 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$).

Finally, using the enhanced umbrella sampling method, the potential of mean force (PMF) for Na^+ and Li^+ permeating through CRCOF-2 and CRCOF-3 membranes was calculated, from which the energy barriers in the processes of ion partitioning and intramembrane diffusion were obtained (Figures 5e and S43). The two peaks in the PMF curves corresponded to the ion partitioning and intramembrane diffusion processes, respectively. As indicated by the peak values, both the partitioning and intramembrane diffusion energy barrier of Na^+ were higher than those of Li^+ , and this tendency was consistent with the results of ion diffusion experiments. Notably, in the CRCOF-2 membrane, the energy barrier for Li^+ during intramembrane diffusion was lower. This phenomenon may be ascribed to the alternating structure of recognition site-charged site-recognition site within the channels of CRCOF-2 membrane. After the ion bound to the crown ether recognition site, the positively charged site without crown ether modification would drive it to move toward the next crown ether recognition site, enabling the hopping transport of ions within the membrane channels and thus reducing the intramembrane diffusion energy barrier (Figure 5f). Within the channels of the CRCOF-3 membrane, as all positively charged

sites were fully occupied by crown ethers, such an alternating structure was absent. Once ions bound to the recognition sites, it was difficult to dissociate, which increased the transport resistance in the membranes.

3 | Conclusion

In summary, this work proposes a crown ether rotaxane-induced construction method to address the challenge of controllably constructing biomimetic membranes with precise ion recognition capability. By regulating the density, type of crown ether, and membrane thickness, the study achieves the synergistic optimization of the pore structure and channel microenvironment in CRCOF nanosheet-stacked membranes, obtaining 1D sub-nanoscale recognition channel membranes. The optimized membrane channels can reduce the energy barrier for target ions to dehydrate and enter the pores, and promote their hopping transport between adjacent recognition sites, thereby achieving highly selective and rapid sieving of target ions. Furthermore, the obtained membranes can maintain high selectivity in mixed salt solutions and retain long-term operational stability during the electrodialysis process, demonstrating practical application value. This research not only provides a design paradigm for advancing the development of biomimetic ion-sieving membranes but also offers a technical pathway for highly selective extraction from complex mixtures. Given the diversity of COFs and the continuous breakthroughs of rotaxane chemistry in the expansion of supramolecular functions, the connection between them in terms of structural customization and performance synergy will become increasingly close, bringing immense opportunities to the large-scale development of biomimetic ion-sieving technology.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (22508303), the International Cooperative Research Program of the National Natural Science Foundation of China (W2512002), the Key Project Program of the National Natural Science Foundation of China (U21A20299), the Tianjin International Joint Laboratory on Belt and Road (24PTLYHZ00270) and the Innovative Research Team of the Chinese University (IRT17R81).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. Y. Xue, Y. Xia, S. Yang, et al., "Atomic-Scale ion Transistor With Ultrahigh Diffusivity," *Science* 372 (2021): 501–503, <https://doi.org/10.1126/science.abb5144>.
2. H. Zhang, X. Li, J. Hou, L. Jiang, and H. Wang, "Angstrom-Scale ion Channels Towards Single-ion Selectivity," *Chemical Society Reviews* 51 (2022): 2224–2254, <https://doi.org/10.1039/D1CS00582K>.
3. H. T. Kratochvil, J. K. Carr, K. Matulef, et al., "Instantaneous ion Configurations in the K^+ ion Channel Selectivity Filter Revealed by 2D

- IR Spectroscopy," *Science* 353 (2016): 1040–1044, <https://doi.org/10.1126/science.aag1447>.
4. W. Xin, J. Fu, Y. Qian, et al., "Biomimetic KcsA Channels With Ultra-Selective K⁺ Transport for Monovalent Ion Sieving," *Nature Communications* 13 (2022): 1701, <https://doi.org/10.1038/s41467-022-29382-6>.
 5. G. Lu, A. Hubao, H. Xu, et al., "Interfacial Supramolecular Interactions Regulated Oligomer Networking Into Robust Sub-Nanochannels for Efficient Osmotic Energy Conversion," *Energy & Environmental Science* 18 (2025): 8515–8526, <https://doi.org/10.1039/D5EE03350K>.
 6. Y. Su, J. Hou, C. Zhao, et al., "Engineering Sub-2 nm Ion-Selective Membranes for Advancing Energy Generation Technologies," *Materials Today* 89 (2025): 206–222, <https://doi.org/10.1016/j.mattod.2025.07.013>.
 7. S. Chang, S. Xue, Y. Yang, J. Pan, and W. Xing, "Porphyrin-Based Metal Organic Frameworks and Covalent Organic Frameworks Materials for Efficient Uranium Extraction," *Coordination Chemistry Reviews* 542 (2025): 216861, <https://doi.org/10.1016/j.ccr.2025.216861>.
 8. B. Mohan, K. Singh, R. K. Gupta, A. J. L. Pombeiro, and P. Ren, "Advanced Materials for Energy Harvesting: Exploring the Potential of MOFs and MXene Membranes in Osmotic Energy Applications," *Progress in Materials Science* 152 (2025): 101457, <https://doi.org/10.1016/j.pmatsci.2025.101457>.
 9. M. Yong, M. Tang, L. Sun, et al., "Sustainable Lithium Extraction and Magnesium Hydroxide co-Production From Salt-Lake Brines," *Nature Sustainability* 7 (2024): 1662–1671, <https://doi.org/10.1038/s41893-024-01435-2>.
 10. D. Jiang, R. Xu, L. Bai, et al., "Insights Into Electrochemical Paradigms for Lithium Extraction: Electrodialysis Versus Capacitive Deionization," *Coordination Chemistry Reviews* 516 (2024): 215923, <https://doi.org/10.1016/j.ccr.2024.215923>.
 11. S. B. Darling, "The Brine of the Times," *Science* 385 (2024): 1421–1422, <https://doi.org/10.1126/science.ads3699>.
 12. X. Tian, C. Ye, L. Zhang, et al., "Enhancing Membrane Materials for Efficient Li Recycling and Recovery," *Advanced Materials* 37 (2025): 2402335, <https://doi.org/10.1002/adma.202402335>.
 13. A. Z. Haddad, L. Hackl, B. Akuzum, G. Pohlman, J. F. Magnan, and R. Kostecki, "How to Make Lithium Extraction Cleaner, Faster and Cheaper — in six Steps," *Nature* 616 (2023): 245–248, <https://doi.org/10.1038/d41586-023-00978-2>.
 14. P. W. Fritz, T. Ashirov, and A. Coskun, "Porous Organic Polymers With Heterocyclic Crown Ethers for Selective Lithium-ion Capture," *Chemistry* 10 (2024): 2207–2219, <https://doi.org/10.1016/j.chempr.2024.03.014>.
 15. G. Wang, L. Shao, and S. Zhang, "Membrane-Ion Interactions Creating Dual-Nanoconfined Channels for Superior Mixed Ion Separations," *Advanced Materials* 37 (2025): 2414898, <https://doi.org/10.1002/adma.202414898>.
 16. X. Liu, P. Liu, H. Wang, and N. M. Khashab, "Advanced Microporous Framework Membranes for Sustainable Separation," *Advanced Materials* 37 (2025): 2500310, <https://doi.org/10.1002/adma.202500310>.
 17. A. Esfandiari, B. Radha, F. C. Wang, et al., "Size Effect in ion Transport Through Angstrom-Scale Slits," *Science* 358 (2017): 511–513, <https://doi.org/10.1126/science.aan5275>.
 18. T. Wu, Y. Qian, Z. Zhu, et al., "Imine-Linked 3D Covalent Organic Framework Membrane Featuring Highly Charged Sub-1 nm Channels for Exceptional Lithium-Ion Sieving," *Advanced Materials* 37 (2025): 2415509, <https://doi.org/10.1002/adma.202415509>.
 19. Y. Liu, Y. Wang, B. Sengupta, et al., "Pillared Laminar Vermiculite Membranes With Tunable Monovalent and Multivalent Ion Selectivity," *Advanced Materials* 37 (2025): 2417994, <https://doi.org/10.1002/adma.202417994>.
 20. N. Bonnet and N. Marzari, "Ion Sieving in Two-Dimensional Membranes from First Principles," *ACS Nano* 19 (2025): 8552–8560.
 21. T. Xu, B. Wu, W. Li, et al., "Perfect Confinement of Crown Ethers in MOF Membrane for Complete Dehydration and Fast Transport of Monovalent Ions," *Science Advances* 10 (2024): eadn0944, <https://doi.org/10.1126/sciadv.adn0944>.
 22. X. Zhao, X. Zhang, X. Xing, et al., "Constructing Angstrom-Level Ion Pocket Array in 1D Channel Wall for Efficient Lithium Ion Sieving," *Advanced Functional Materials* 35 (2025): 2416628, <https://doi.org/10.1002/adfm.202416628>.
 23. D. Wang, Q. Chen, and J. Wang, "Constructing High-Selective Cation Exchange Membrane via Embedding Sulfonated Lithium-Ion Sieve for Enhanced Lithium Recycling From Spent Batteries Effluents," *Advanced Functional Materials* 35 (2025): 2425819, <https://doi.org/10.1002/adfm.202425819>.
 24. M. Wang, T. Ma, Z. Wu, et al., "Construction of Local-Ion Trap in Phase-Reversed Mixed Matrix COF Membranes for Ultrahigh Ion Selectivity," *Angewandte Chemie International Edition* 64 (2025): e202504990, <https://doi.org/10.1002/anie.202504990>.
 25. X. Pang, B. Shi, Y. Liu, et al., "Phosphorylated Covalent Organic Framework Membranes Toward Ultrafast Single Lithium-Ion Transport," *Advanced Materials* 36 (2024): 2413022, <https://doi.org/10.1002/adma.202413022>.
 26. D. W. Burke, Z. Jiang, A. G. Livingston, and W. R. Dichtel, "2D Covalent Organic Framework Membranes for Liquid-Phase Molecular Separations: State of the Field, Common Pitfalls, and Future Opportunities," *Advanced Materials* 36 (2024): 2300525, <https://doi.org/10.1002/adma.202300525>.
 27. C. Qian, L. Feng, W. L. Teo, et al., "Imine and Imine-Derived Linkages in Two-Dimensional Covalent Organic Frameworks," *Nature Reviews Chemistry* 6 (2022): 881–898, <https://doi.org/10.1038/s41570-022-00437-y>.
 28. C. Qian and X. Zhao, "Two-Dimensional Heteropore Covalent Organic Frameworks: From Construction to Functions," *Accounts of Chemical Research* 58 (2025): 1192–1209, <https://doi.org/10.1021/acs.accounts.4c00799>.
 29. J. Han, Z. Xing, Q. Guo, et al., "Covalent Organic Framework Nanosheets for the Assembly of Efficient Membrane Bioreactors," *Angewandte Chemie International Edition* 64 (2025): e202505797.
 30. D. Wu, Z. Xing, Q. Guo, et al., "Engineering Bipolar Covalent Organic Framework Membranes for Selective Acid Extraction," *Angewandte Chemie International Edition* 64 (2025): e202503945.
 31. X. Liu, W. Lin, K. Bader Al Mohawes, and N. M. Khashab, "Ultrahigh Proton Selectivity by Assembled Cationic Covalent Organic Framework Nanosheets," *Angewandte Chemie International Edition* 64 (2025): e202419034.
 32. X. Wu, X. Han, Y. Liu, Y. Liu, and Y. Cui, "Control Interlayer Stacking and Chemical Stability of Two-Dimensional Covalent Organic Frameworks via Steric Tuning," *Journal of the American Chemical Society* 140 (2018): 16124–16133, <https://doi.org/10.1021/jacs.8b08452>.
 33. X. Tian, H. Huan, K. Zhang, et al., "Building Unit Engineering Toward COF Membranes With Controlled Stacking for H₂ Purification," *Advanced Materials* 37 (2025): 2504622, <https://doi.org/10.1002/adma.202504622>.
 34. L. Ren, H. Wu, J. Han, and J. Chen, "Rotaxane Mediated Covalent Organic Framework Membranes for Efficient Desalination," *Advanced Functional Materials* 35 (2025): 2425244, <https://doi.org/10.1002/adfm.202425244>.
 35. T. Huang, H. Jiang, J. C. Douglas, et al., "Single Solution-Phase Synthesis of Charged Covalent Organic Framework Nanosheets With High Volume Yield," *Angewandte Chemie International Edition* 62 (2023): e202209306, <https://doi.org/10.1002/anie.202209306>.
 36. B. Wang, S. Zhang, Z. Liu, et al., "Noncovalent Complex Modulated Fabrication of COF Membrane for Organic Solvent Nanofiltration," *Small* 21 (2025): 2500927, <https://doi.org/10.1002/smll.202500927>.

37. S. Zheng, S. Bi, Y. Fu, et al., "3D Crown Ether Covalent Organic Framework as Interphase Layer Toward High-Performance Lithium Metal Batteries," *Advanced Materials* 36 (2024): 2313076, <https://doi.org/10.1002/adma.202313076>.
38. X. Ruan, Y. Yang, W. Liu, et al., "Mechanical Bond Approach to Introducing Self-Adaptive Active Sites in Covalent Organic Frameworks for Zinc-Catalyzed Organophosphorus Degradation," *ACS Central Science* 7 (2021): 1698–1706, <https://doi.org/10.1021/acscentsci.1c00941>.
39. A. Llamosi, M. P. Szymański, and A. Szumna, "Molecular Vessels From Preorganised Natural Building Blocks," *Chemical Society Reviews* 53 (2024): 4434–4462, <https://doi.org/10.1039/D3CS00801K>.
40. T. Lu and F. Chen, "Multiwfn: A Multifunctional Wavefunction Analyzer," *Journal of Computational Chemistry* 33 (2012): 580–592, <https://doi.org/10.1002/jcc.22885>.
41. W. Jiang, J. Zhou, X. Zhong, et al., "Axial Alignment of Covalent Organic Framework Membranes for Giant Osmotic Energy Harvesting," *Nature Sustainability* 8 (2025): 446–455, <https://doi.org/10.1038/s41893-024-01493-6>.
42. S. E. Neumann, J. Kwon, C. Gropp, et al., "The Propensity for Covalent Organic Frameworks to Template Polymer Entanglement," *Science* 383 (2024): 1337–1343, <https://doi.org/10.1126/science.adf2573>.
43. X. Tian, L. Cao, K. Zhang, et al., "Molecular Weaving Towards Flexible Covalent Organic Framework Membranes for Efficient Gas Separations," *Angewandte Chemie International Edition* 64 (2025): e202416864, <https://doi.org/10.1002/anie.202416864>.
44. X. Ruan, C. Zhang, Y. Zhu, et al., "Constructing Mechanical Shuttles in a Three-dimensional (3D) Porous Architecture for Selective Transport of Lithium Ions," *Angewandte Chemie International Edition* 62 (2023): e202216549, <https://doi.org/10.1002/anie.202216549>.
45. P. P. Power and X. Xiaojie, "Mononuclear Lithium Amides Using a Crown Ether: X-Ray Crystal Structure of Lithium Bis(Trimethylsilyl)Amide-12-Crown-4, [Li(sime 3) 2 ·12-Crown-4]," *Journal of the Chemical Society, Chemical Communications* 6 (1984): 358–359, <https://doi.org/10.1039/C39840000358>.
46. D. Jiang, J. P. Hill, J. Henzie, et al., "Selective Electrochemical Capture of Monovalent Cations Using Crown Ether-Functionalized COFs," *Journal of the American Chemical Society* 147 (2025): 12460–12468, <https://doi.org/10.1021/jacs.4c16346>.
47. D. Yang, Y. Yang, T. Wong, et al., "Solution-Processable Polymer Membranes With Hydrophilic Subnanometre Pores for Sustainable Lithium Extraction," *National Water* 3 (2025): 319–333, <https://doi.org/10.1038/s44221-025-00398-8>.
48. Y. Guo, J. He, J. Zhang, et al., "Deciphering co-ion and counterion transport in polyamide desalination membranes reveals ion selectivity mechanisms," *Science Advances* 11 (2025): eadu8302, <https://doi.org/10.1126/sciadv.adu8302>.
49. R. Xu, H. Yu, J. Ren, et al., "Regulate Ion Transport in Subnanochannel Membranes by Ion-Pairing," *Journal of the American Chemical Society* 147 (2025): 17144–17151, <https://doi.org/10.1021/jacs.5c02722>.
50. Q. Meng, J. Li, Z. Lai, et al., "Optimizing Selectivity via Membrane Molecular Packing Manipulation for Simultaneous Cation and Anion Screening," *Science Advances* 2024, 10, eado8658.
51. T. Ye, H. Gao, Q. Li, et al., "Highly Selective Lithium Transport Through Crown Ether Pillared Angstrom Channels," *Angewandte Chemie International Edition* 63 (2024): e202316161, <https://doi.org/10.1002/anie.202316161>.
52. X. Dong, X. Ai, W. He, et al., "Heteropolyacid Ligands in Two-Dimensional Channels Enable Lithium Separation From Monovalent Cations," *ACS Nano* 19 (2025): 4233–4241, <https://doi.org/10.1021/acsnano.4c10606>.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File 1: The authors have cited additional references within the Supporting Information.